

Amodal3R

Amodal 3D Reconstruction from Occluded 2D Images

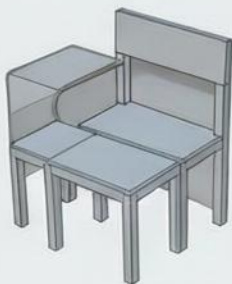
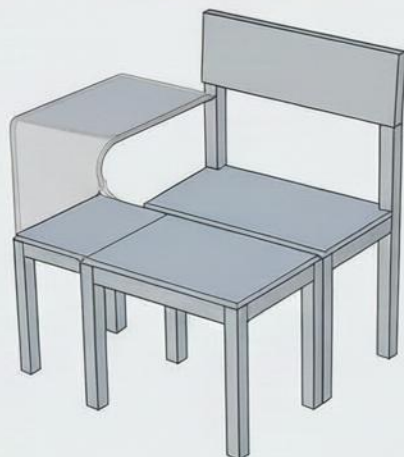


从遮挡图像到完整 3D · Single Image → Complete 3D

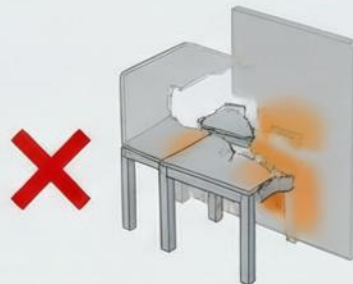
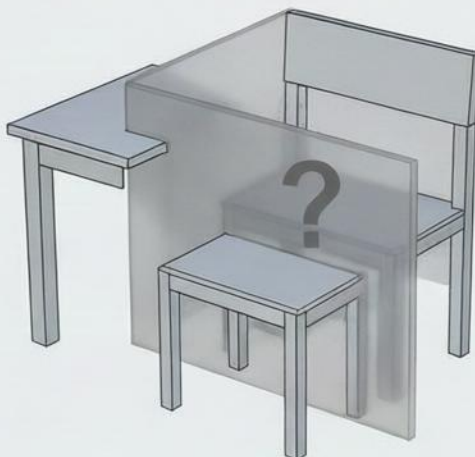
The Occlusion Problem — 遮挡难题

Why Occlusion Matters / 为什么遮挡是难题

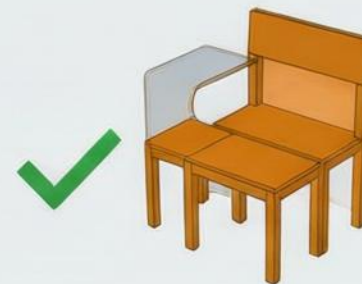
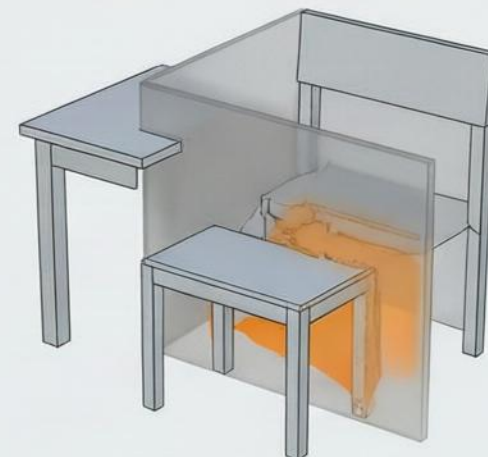
Complete View



Occluded View



Amodal 3R (Ours)



Occlusion is everywhere. Standard methods fail. Amodal understanding is necessary.
遮挡无处不在，传统方法失败，非模态理解不可或缺。

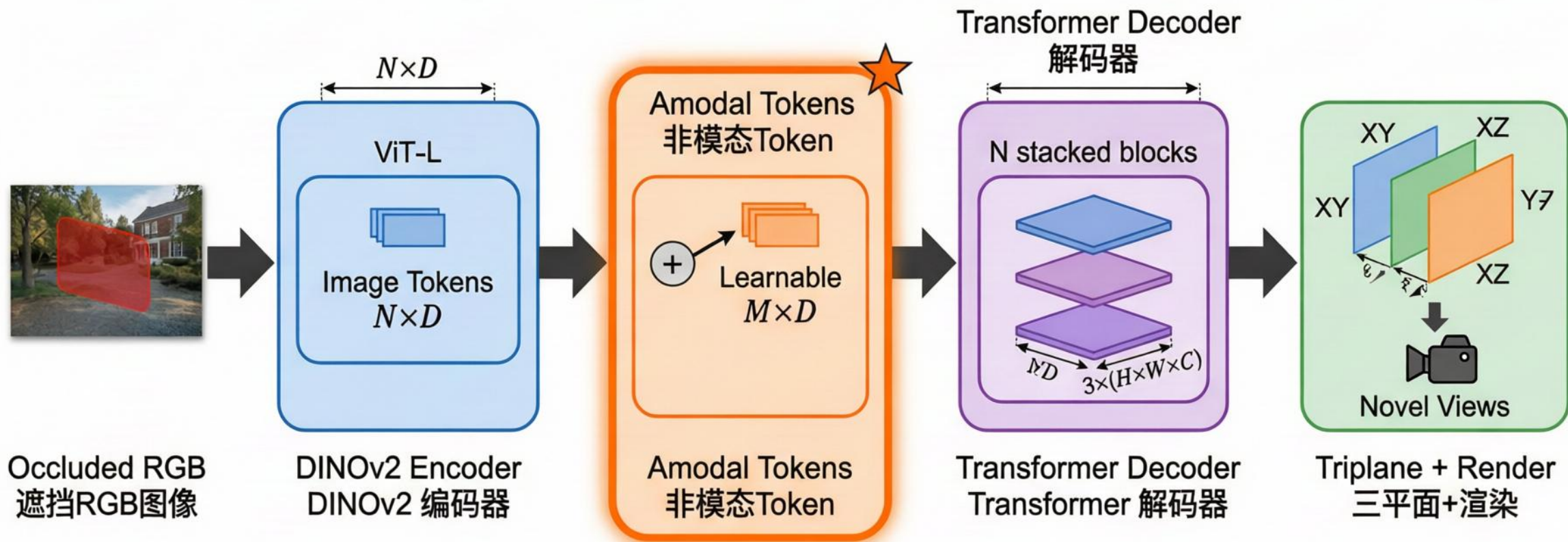
Our Approach — 本文方案

Core Idea: Amodal Tokens / 核心思想: 非模态Token



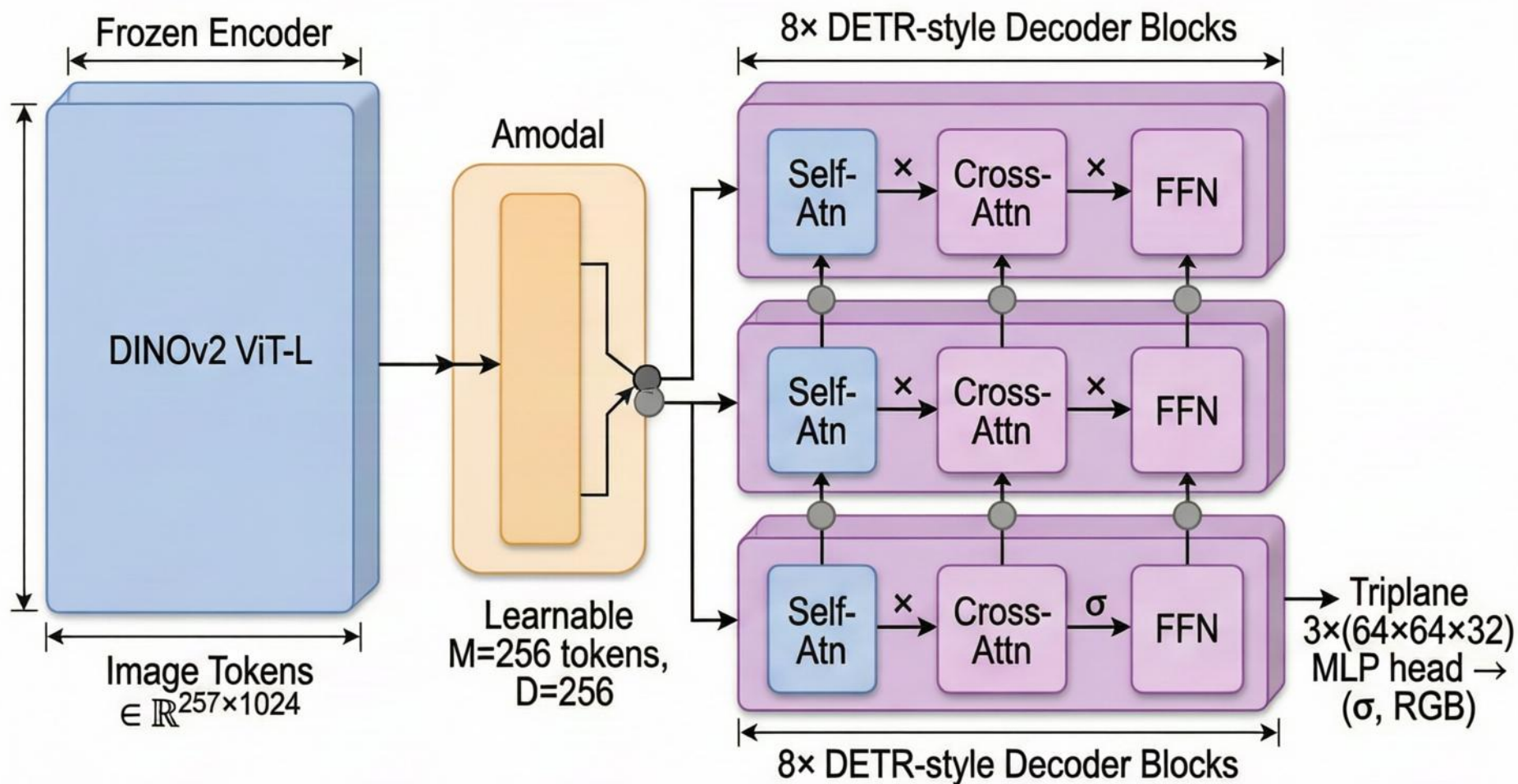
Learnable embeddings that explicitly model occluded 3D geometry
可学习嵌入，显式建模被遮挡的 3D 几何

Pipeline Overview — 方法总览



Single Image → Encoder → Amodal Tokens → Decoder → Triplane → Render / 单图→编码→Token→解码→三平面→渲染

Architecture Details — 架构细节



Encoder
DINOv2 ViT-L, frozen

Image tokens
 257×1024

Amodal tokens
 256×256

Decoder layers
8

Triplane resolution
 $64 \times 64, 32$ channels

Rendering
64 samples per ray

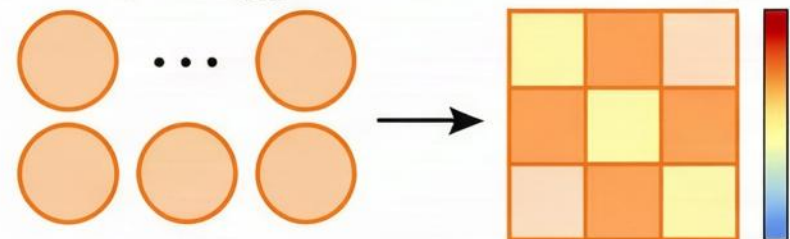
DINOv2 (frozen) provides robust visual features; Decoder learns to complete 3D geometry via Amodal Tokens

Amodal Token Mechanism / 非模态Token机制

M amodal tokens



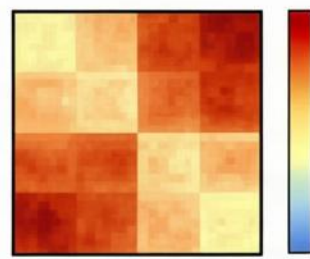
$T_1 \dots T_M$ Linear Q



$T_1 \dots T_M$

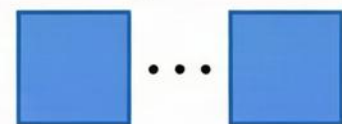
Query Q

$Q \times K^T$
Scale
($1/\sqrt{d}$)



Attention
Weights

N image tokens

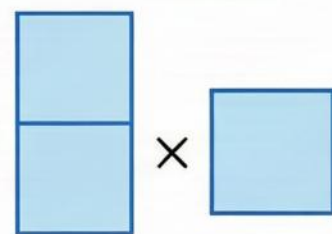


$I_1 \dots I_N$

Linear K , Linear V



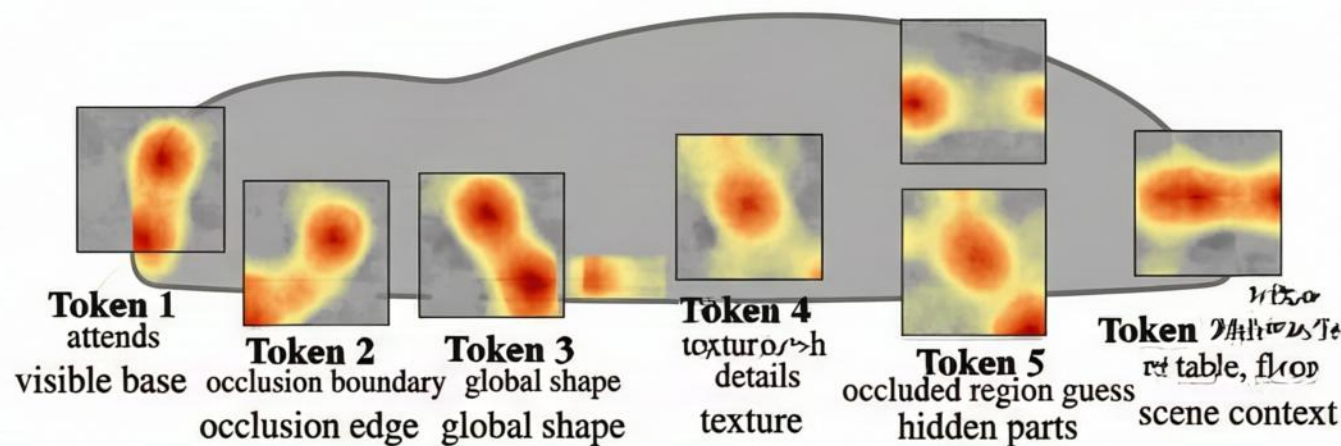
Key K Value V



$\times V$

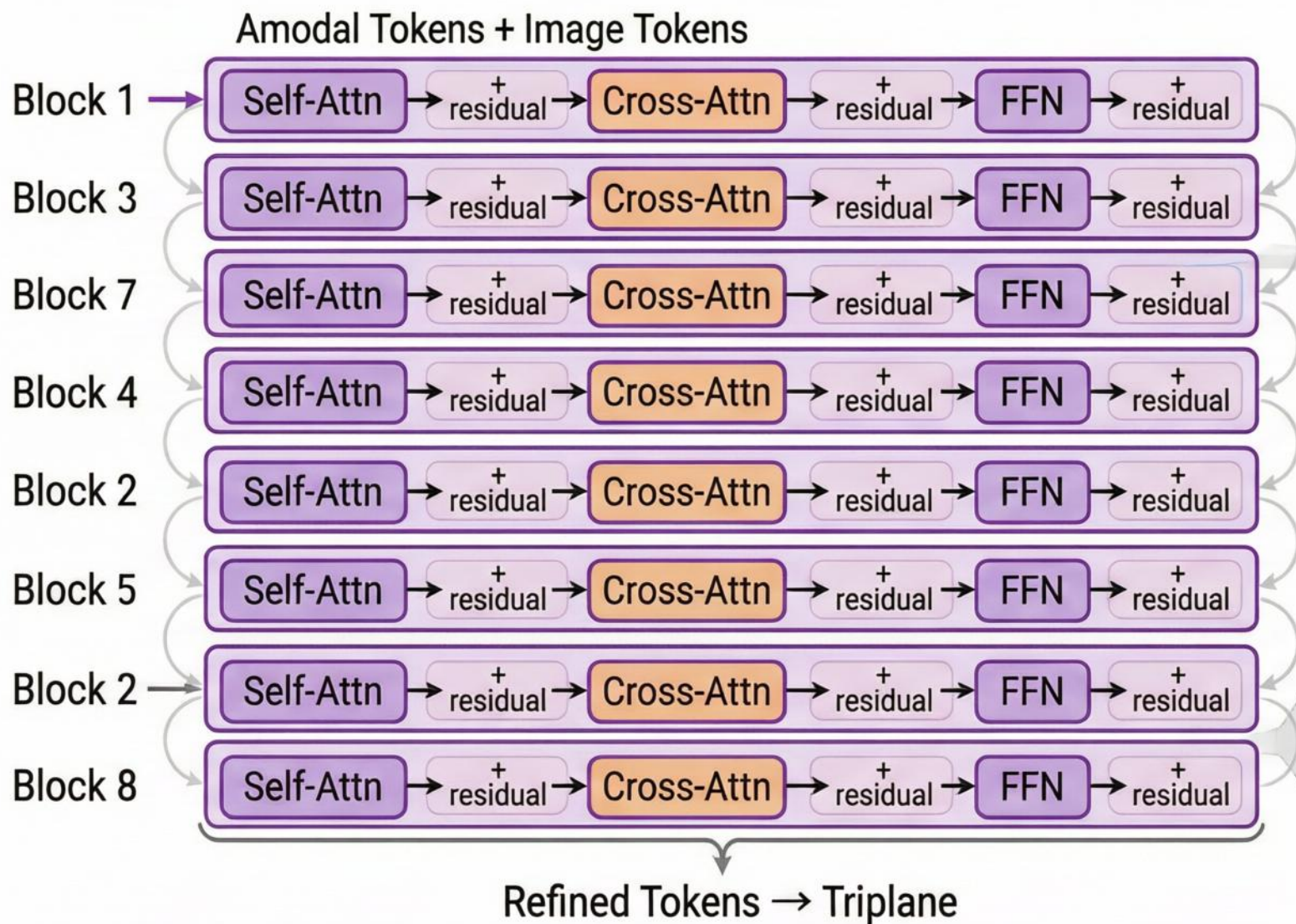


Output Tokens

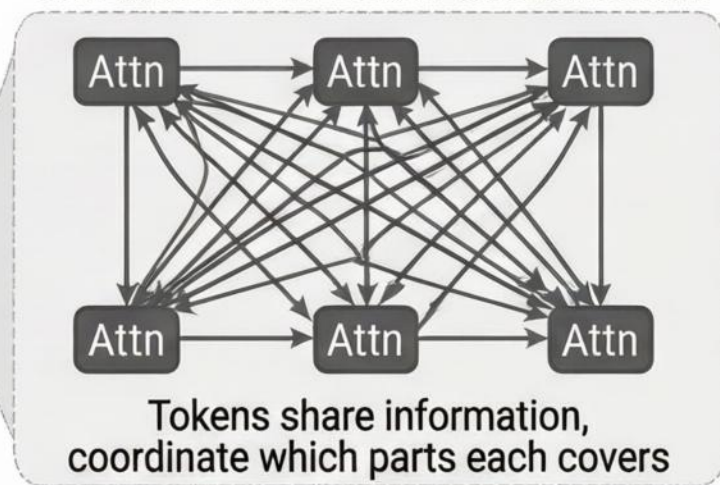


Token 各司其职：不同 Token 学会询问不同的几何线索

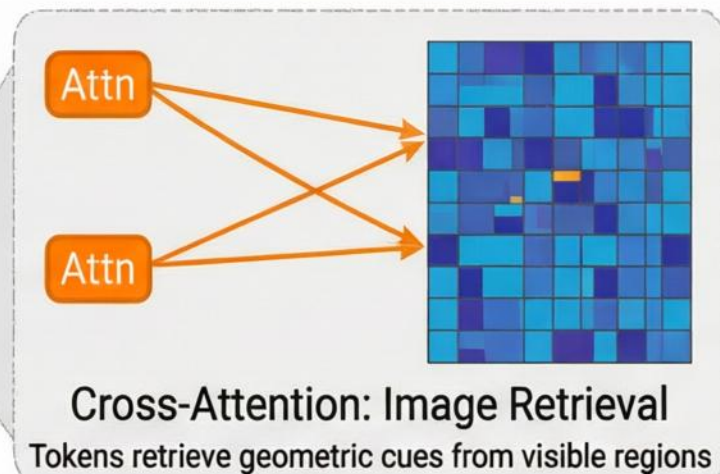
Transformer Decoder / Transformer 解码器内部



Self-Attention: Token Coordination



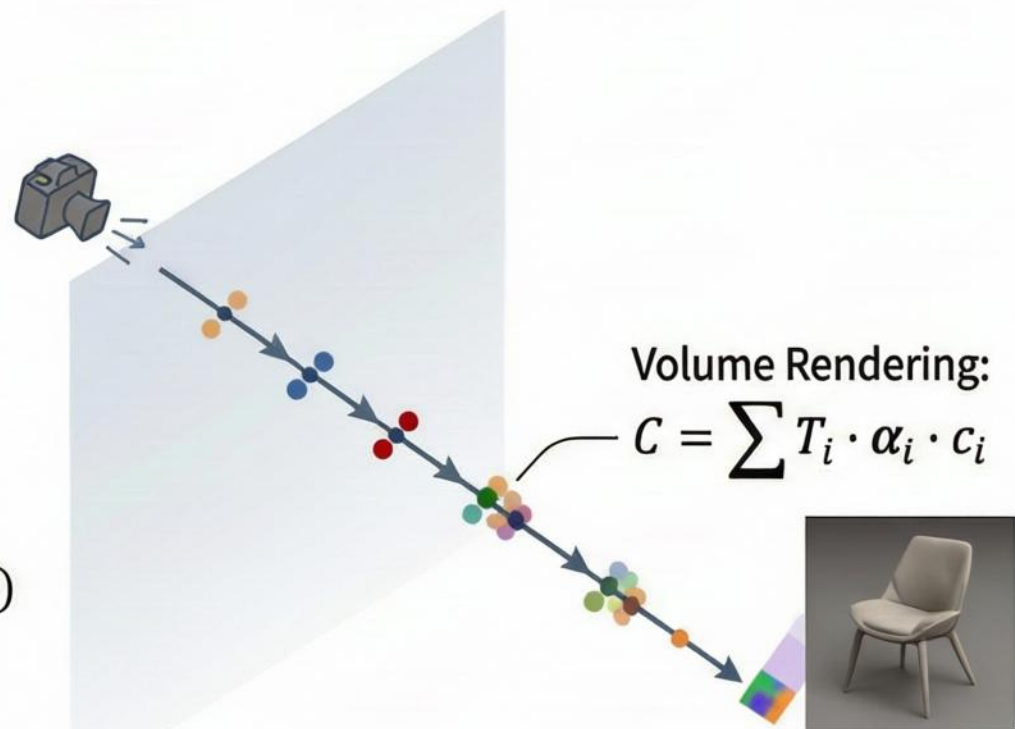
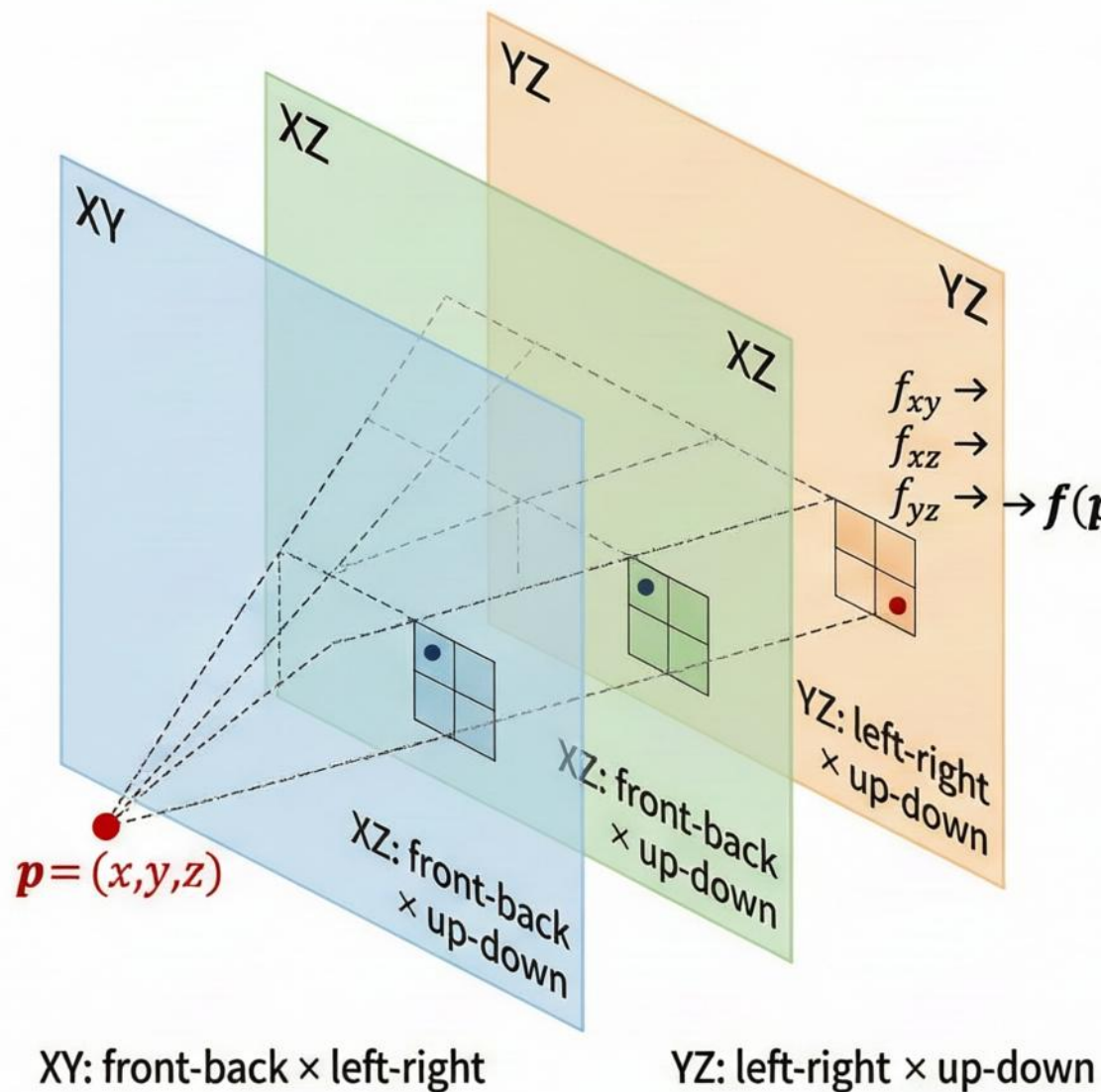
Token 之间协调分工



Token 从可见区域检索几何线索

8 blocks. Self-attn for coordination. Cross-attn for geometric retrieval. 8层解码器: Self-attn 协调, Cross-attn 检索

Triplane & Representation & Volume Rendering / 三平面表示与体渲染



Triplane = efficient 3D representation.

Any 3D point $\rightarrow$ project + interpolate + sum $\rightarrow$ feature.
三平面：高效 3D 表示，任意点 $\rightarrow$ 投影+插值+求和 $\rightarrow$ 特征

Training Strategy — 训练策略

role: method journal-minisert style: journal-minimal language: bilingual 11/2/23

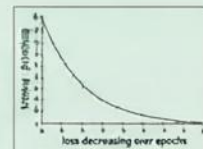
Stage 1: Occlusion Simulation / 遮挡模拟



Stage 2: Multi-View Rendering Loss / 多视图渲染损失



$$L_{\text{total}} = L_{\text{rgb}} + \lambda_{\text{mask}} \cdot L_{\text{mask}} + \lambda_{\text{percep}} \cdot L_{\text{LPIPS}}$$



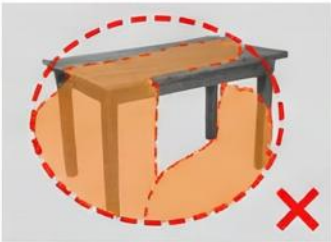
Trained on synthetic occlusions only. Zero-shot generalizes to real occluded images. / 仅用合成遮挡训练，零样本泛化到真实遮挡

Key Results — 关键实验结果

Method	PSNR↑	SSIM↑	SSIM↑	LPIPS↓	Amodal-PSNR↑
Baseline A	↑ X.XX	⇕ X.XX	↑ X.XX	↑ X.XX	↑ X.XX
Baseline B	X.XX	X.XX	X.XX	X.XX	X.XX
	X.XX	↑	↑	X.XX	↑
	X.XX	X.XX	X.XX	X.XX	X.XX
Baseline C	⇕ X.XX	⇕ X.XX	⇕ X.XX	⇕ X.XX	⇕ X.XX
Amodal3R (Ours)	✓ ✗	+3.2 dB PSNR on occluded regions →			✓ ✗



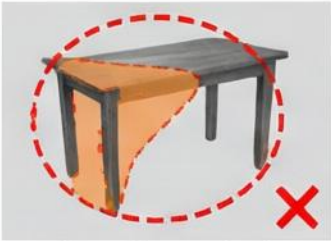
Input



Baseline output



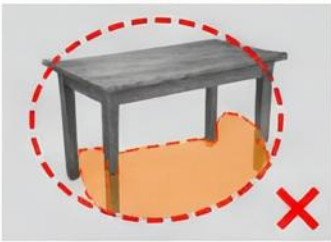
Input



Baseline output



Input



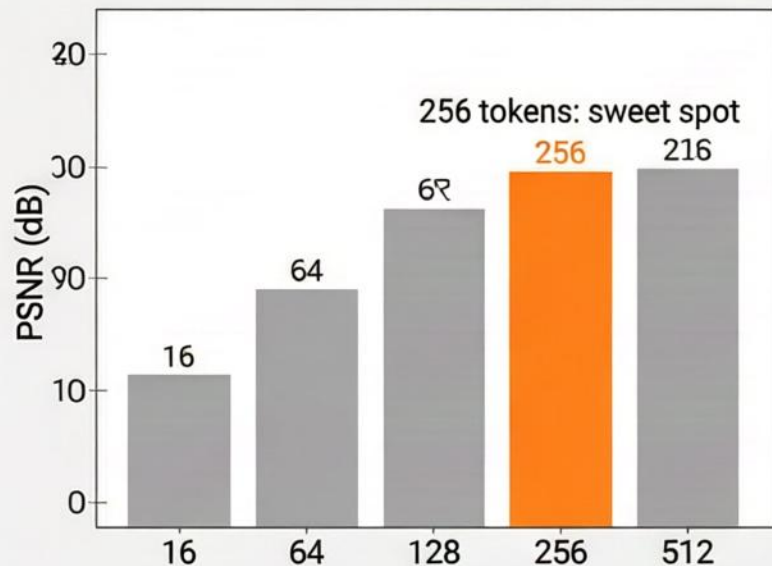
Amodal3R output



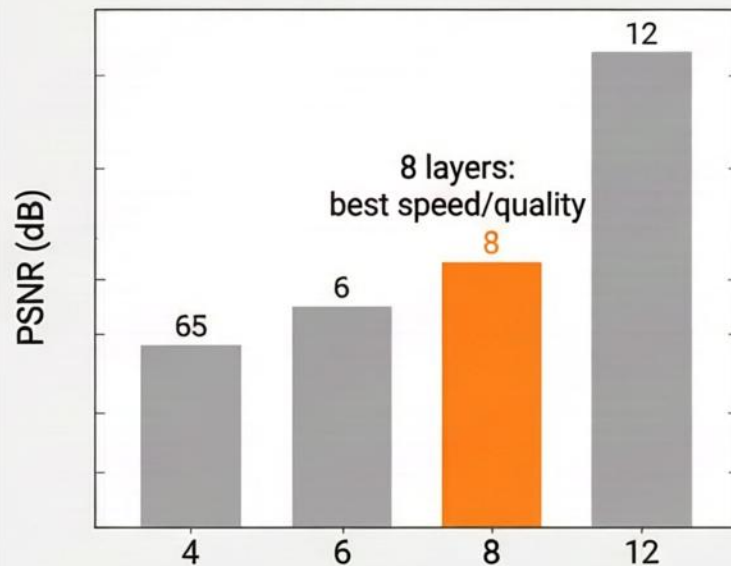
Amodal3R consistently outperforms baselines, especially on heavily occluded regions
在所有指标上超越基线，遮挡区域优势显著

Ablation Study — 消融实验

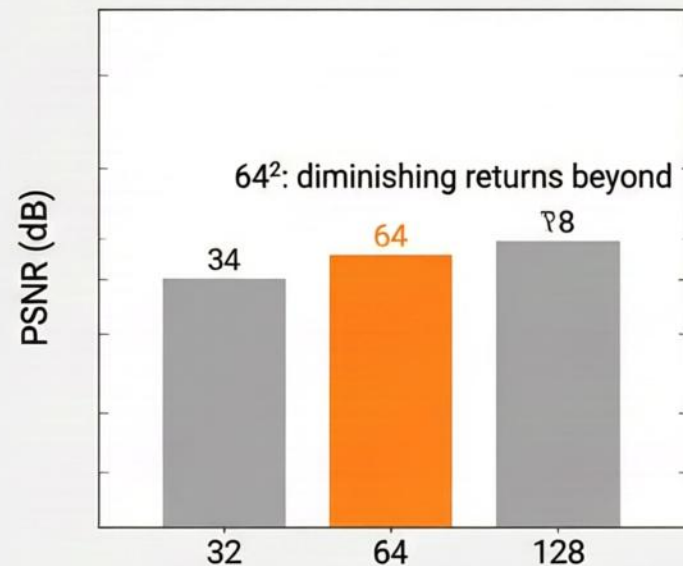
Amodal Token Count / Token 数量



Decoder Layers / 解码器层数



Triplane Resolution / 三平面分辨率



w/o Amodal Tokens	PSNR	Amodal-PSNR
w/o Cross-Attn		
w/o Triplane		
Full Model		

All components contribute. Amodal Tokens provide the largest gain.
所有组件均有贡献，Amodal Token 收益最大

Real-World Generalization – 真实场景泛化

Chair behind
桌子后的椅子



Car behind
树后的汽车



Sofa behind
植物后的沙发



- Trained on:
Synthetic occlusions only
仅合成遮挡训练
- Tested on: Real in-the-wild photos
真实场景照片测试
- Zero-shot transfer —
no fine-tuning needed
零样本迁移，无需微调

Zero-Shot ✓

Zero-Shot ✓

Synthetic training → Real-world generalization. The model learns a general amodal understanding.
合成训练→真实泛化，模型学到了通用的非模态理解能力

Summary & Takeaways — 总结



Amodal Tokens 非模态Token

Learnable tokens that explicitly model occluded 3D geometry via cross-attention

可学习Token，通过交叉注意力显式建模被遮挡的3D几何



Feed-Forward Efficiency 前馈效率

Single image → Complete 3D in one forward pass.
No multi-view, no optimization.

单图→完整3D，一次前馈，无需多视图或迭代优化



State-of-the-Art Quality 最高质量

Outperforms all baselines.
+3.2dB on occluded regions.
Generalizes to real scenes.

超越所有基线，遮挡区域+3.2dB，泛化到真实场景

Amodal3R: What you see is not all you get — we reconstruct what's hidden.
你看到的不是全部——我们重建被隐藏的部分。

Code & Paper: github.com/... / arXiv 2503.13439